Hriday Adani

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Education

Rutgers University - New Brunswick — New Brunswick, NJ
Bachelor of Science; Computer Science and Economics — *Expected Graduation - May 2027*

Experience

Scientific Software Research Assistant — Jul. 2026 - Present
RCSB Protein Data Bank — *New Brunswick, NJ*

- Orchestrated a new MCP server tool exposing PDB search API to LLM agents enabling multi-service AND/OR queries in a single call, by composing the filter nodes from the 7 extended tools through a custom validation layer
- Fixed **2 silent data corruption bugs** in MCP server's result-grouping logic by tracing parameter flow against the tool's own docstrings, preventing the formation of invalid API parameters and user configurations

Summer Research Fellow (CSI SURF) — May. 2026 - Jul. 2026
Rutgers Artificial Intelligence and Data Science — *New Brunswick, NJ*

- Redesigned a dictionary-driven column-typing classifier in py-mmcif, an open-source Python package, cutting molecular structure file sizes by **27%** across an archive serving ∼**5M downloads** per day; shipped to production
- Optimized floating-point binary encoding in py-mmcif's compression pipeline, cutting compressed file sizes by up to **76%** across 37 varied PDB structures, working with a team of 4 researchers on the encoding chain design

Projects

StorageKV | *Rust (zero runtime dependencies)*

- Engineered an embedded RocksDB/LevelDB-inspired key-value storage engine from scratch on an LSM-tree architecture with a WAL, immutable sorted SSTables with sparse indexing, and size-tiered compaction
- Implemented crash recovery via manifest checkpoint plus WAL replay, enforcing fsync ordering and atomic file installation so any acknowledged write survives crashes in the write path, similar to PostgreSQL's architecture
- Measured **sub-12μs p99** reads and quantified the cost of durability at approximately **1000x** over an in-memory baseline by isolating fsync overhead in same-machine benchmarks

ClauseWatch | *Next.js, FastAPI, PostgreSQL, OpenAI*

- Extracted structured fields from PDF and DOCX contracts at **97.9%** field-level accuracy (100% on effective and renewal dates) across a validated 30-document test set, using confidence-based OpenAI structured outputs
- Engineered 5 independently-secured HMAC token systems including webhooks, downloads, password reset, email verification, and cron auth, each with a dedicated secret and constant-time comparison to prevent timing attacks
- Designed a multi-tenant data isolation architecture by enforcing tenant-scoped queries at the ORM layer across every API endpoint, verified via 6 automated integration tests confirming **zero cross-tenant data leakage**

AccessForge | *FastAPI, LangGraph, PostgreSQL, Redis/RQ, Docker, Playwright, Axe-core*

- Developed an end-to-end accessibility remediation platform that scans React apps, generates and independently verifies WCAG fixes, and opens GitHub PRs, designed as a deterministic state machine using LangGraph
- Architected a crash-recoverable async pipeline using PostgreSQL-backed state and Redis workers, verified via a real worker crash captured during development, resuming task with **0 duplicate GitHub branches** or lost progress
- Devised a confidence-gated localization check that deferred **68%** of detected violations to human review instead of guessing when no signal matched a source file, resolving **100%** of attempted accessible-name fixes on first pass

UniHop - Winner HackRU Fall 2026 | *Java, SpringBoot, PostgreSQL, GitHub Actions*

- Built a Rutgers-Only campus rideshare mobile app with a 3-person team in 24 hours, matching riders to the nearest driver through 15-sec realtime offers that redispatch on decline, with PIN-verified trips and fixed rates under $10
- Eliminated partial rides and double charges by building an idempotent SpringBoot + PostgreSQL payment layer that authorizes within the ride-creation transaction, then captures on completion or voids on cancellation
- Gated every merge to main with GitHub Actions CI running **210 backend tests**, the frontend build, and an end-to-end smoke test of the JAR against disposable PostgreSQL, blocking 3 breaking Dependabot upgrades

Technical Skills

Languages: Java, Python, C, PHP, JavaScript (React, Next), SQL, HTML/CSS, Rust
Libraries/Frameworks: Tailwind CSS, Flask, FastAPI, LangGraph, Playwright, Axe-core, SpringBoot
Developer Tools: VS Code, Claude Code, Git, Docker, WSL, Alembic, Neon, Cloudflare, MCP, Redis